Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

A bookstore wants to analyze the titles of books to determine their longest word in each title. This helps in designing banners and covers.

Your task is to write a program that, given a sentence (book title), finds and prints the longest word. If multiple words have the same maximum length, print the first one.

Input Format

The input contais a single line containing a sentence representing the book title.

Output Format

The output prints a string representing the longest word in the sentence (book title).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The Chronicles of Narnia
Output: Chronicles

Answer

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Read the input sentence (book title)
        String sentence = scanner.nextLine();
        scanner.close();

        // Split sentence into words
        String[] words = sentence.split(" ");
        String longestWord = "";
        int maxLength = 0;

        for (String word : words) {
            if (word.length() > maxLength) {
                longestWord = word;
                maxLength = word.length();
            }
        }

        // Print the longest word
        System.out.println(longestWord);
    }
}
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

In a college, students are required to create unique usernames for accessing the digital library.

The librarian needs your help to verify whether the usernames entered by students are valid.

A username is considered valid if:

It contains only letters (a–z, A–Z) and digits (0–9).Its length is between 5 and 15 characters (inclusive).It must start with a letter (not a digit).

Your task is to determine whether each username in the list is valid or not.

Input Format

The first line of input contains an integer T, representing the number of usernames to check.

The next T lines each contain a string S, representing a username.

Output Format

For each username S, the output print "YES" if it is valid.

Otherwise, the output print "NO".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1
Alice123

Output: YES

Answer

```
import java.util.*;

public class Main{
  public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);

        // Read number of usernames
        int T = Integer.parseInt(scanner.nextLine());

        for (int i = 0; i < T; i++) {
            String username = scanner.nextLine();
            if (isValidUsername(username)) {
                System.out.println("YES");
            } else {
                System.out.println("NO");
            }
        }

        scanner.close();
    }

    // Validation method
    private static boolean isValidUsername(String username) {
        // Check length constraint
        if (username.length() < 5 || username.length() > 15) {
            return false;
        }

        // Check that it starts with a letter
        if (!Character.isLetter(username.charAt(0))) {
            return false;
        }

        // Check that all characters are letters or digits
        for (char ch : username.toCharArray()) {
            if (!Character.isLetterOrDigit(ch)) {
                return false;
            }
        }

        return true; // All checks passed
    }
}
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Meera is practicing her English vocabulary. She wants to focus on words that have more vowels in them, as they help improve her pronunciation. She decides to extract only those words from a sentence that contain at least two vowels.

Your task is to help Meera by writing a program that finds such words from the given sentence.

Input Format

The input contains a string representing the sentence.

Output Format

The output prints all the words that contain at least two vowels, separated by a space.

If no such word exists, print "No words with two vowels".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: This is an example sentence
Output: example sentence

Answer

```
import java.util.*;

public class Main {
   public static void main(String[] args) {
      Scanner scanner = new Scanner(System.in);

      String sentence = scanner.nextLine();
      scanner.close();

      String[] words = sentence.split(" ");
      List<String> result = new ArrayList<>();
```

```
        for (String word : words) {
            if (countVowels(word) >= 2) {
                result.add(word);
            }
        }

        if (result.isEmpty()) {
            System.out.println("No words with two vowels");
        } else {
            for (String w : result) {
                System.out.print(w + " ");
            }
        }
    }

    private static int countVowels(String word) {
        int count = 0;
        String vowels = "aeiouAEIOU";

        for (char ch : word.toCharArray()) {
            if (vowels.indexOf(ch) != -1) {
                count++;
            }
        }

        return count;
    }
}
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

Riya is preparing for a vocabulary test. Her teacher told her to focus on long words in her practice sentences, specifically words that have at least 5 letters.

Riya wants to write a program that will help her identify such words quickly.

Your task is to help Riya by printing all the words in a given sentence that have a length greater than or equal to 5.

If no such word exists, display "No long words found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words having length ≥ 5, separated by a space.

If no such word is found, print "No long words found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The quick brown fox jumps over the lazy dog
Output: quick brown jumps

Answer

```
import java.util.*;

public class Main {
  public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    String sentence = scanner.nextLine();
    scanner.close();

    String[] words = sentence.split(" ");
    List<String> longWords = new ArrayList<>();

    for (String word : words) {
      if (word.length() >= 5) {
        longWords.add(word);
      }
    }
```

```
        if (longWords.isEmpty()) {
            System.out.println("No long words found");
        } else {
            for (String w : longWords) {
                System.out.print(w + " ");
            }
        }
    }
}
```

Status : Correct ***Marks : 10/10***